

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

BU Spark!

September 2024 – December 2024

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

Boston University Engineering IT

January 2024 – July 2025

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow*

May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows
- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

Scalable Chat Platform | *Go, AWS (ECS, RDS, ElastiCache, ALB), WebSockets, Docker*

[Month] 2026

- Architected cloud-native messaging platform on AWS ECS with auto-scaling policies, PostgreSQL RDS, and Redis ElastiCache, handling [X] concurrent WebSocket connections (load test with Artillery.js)
- Implemented horizontal scaling with Application Load Balancer and sticky sessions, deploying via GitHub Actions CI/CD with blue-green deployment strategy achieving [X]% uptime (CloudWatch metrics)
- Integrated comprehensive observability with CloudWatch dashboards, custom metrics, and structured logging, enabling sub-[X]ms P95 response time monitoring (APM traces and database query analysis)

Audio Transcription Platform | *Java Spring Boot, Python FastAPI, React, PostgreSQL*

September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling
- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

TECHNICAL SKILLS

Security Languages: Solidity, Circom, Java, Python, JavaScript, TypeScript, SQL (PostgreSQL)

Cryptographic Protocols: Zero-Knowledge Proofs, Groth16 Protocol, SHA256 Hash Verification, Privacy-Preserving Verification

Smart Contract Security: Solidity, Ethereum, ERC-20 Tokens, Transaction Management, Economic Incentive Mechanisms

Infrastructure Security: AWS Security, Application Load Balancer, Authentication Systems, Access Control, Security Middleware

Authentication & Authorization: Auth0, MetaMask Integration, User Account Management, Permission Systems, Security Audits

Endpoint & Network Security: CrowdStrike, Network Security, Incident Response, Security Monitoring, Zero Security Incidents